Separation of fast and slow oscillations.

Consider free particle subject to oscillating force

$$m\ddot{x} = F\cos\omega t \Rightarrow x(t) = \underbrace{A + Bt}_{\substack{\text{solution of} \\ \text{homogeneous} \\ \text{equ. } \ddot{x}=0}} - \frac{F\cos\omega t}{\omega^2}$$

Observe that as $\omega \to \infty$ correction to the free motion becomes very weak. This gives an idea of separating "slow" and "fast" motions when oscillating frequency is sufficiently large.

So consider $m\ddot{x} = -\frac{dU}{dx} + f(x,t)$

↑ high frequency force

$$f(x,t) = f_1(x)\cos\omega t + f_2(x)\sin\omega t$$

let $x(t) = \underset{\text{slow}}{X(t)} + \underset{\text{fast}}{\xi(t)}$

Assume: $X(t)$, being slow, changes very little over the period $T = \frac{2\pi}{\omega}$ of $\xi(t)$ motion.

We also expect the fast component to be small: $\xi \ll X$

$$\Rightarrow U(x) = U(X+\xi) \approx U(X) + \xi\frac{dU}{dX} + \dots$$

$$f(x,t) = f(X+\xi,t) = f(X,t) + \xi\frac{\partial f}{\partial X} + \dots$$

$$\Rightarrow m(\ddot{X} + \ddot{\xi}) = \underbrace{-\frac{dU}{dX}}_{\text{slow term}} \underbrace{- \xi\frac{d^2U}{dX^2} + f(X,t) + \xi\frac{\partial f}{\partial X}}_{\text{fast terms}} \quad (1)$$

$$\Rightarrow \quad m\ddot{\xi} = f(X,t) - \xi \frac{d^2U}{dX^2} + \xi \frac{df}{dX}$$

Since ξ is small, these terms can be neglected

$$m\ddot{\xi} = f(X,t) = f_1(X)\cos\omega t + f_2(X)\sin\omega t$$

Note that $X = X(t)$ but it is slow on the scale of $\frac{2\pi}{\omega}$ $\Rightarrow$ can treat $X(t)$ as a constant here!

$$\Rightarrow \quad \xi(t) = -\frac{f_1(X)\cos\omega t + f_2(X)\sin\omega t}{m\omega^2} = -\frac{f(X,t)}{m\omega^2}$$

Now go back to (1) and integrate both sides $\int_0^{2\pi/\omega} dt$:

$$\frac{1}{T}\underbrace{\int_0^{2\pi/\omega} dt\,(m\ddot{X} + m\ddot{\xi})}_{m\ddot{X}} = \frac{1}{T}\int_0^{2\pi/\omega} dt\left(-\frac{dU}{dX} - \xi\frac{d^2U}{dX^2} + f(X,t) + \xi\frac{df}{dX}\right)$$

$$m\ddot{X} = -\frac{dU}{dX} + \frac{1}{m\omega^2}\underbrace{\int_0^{2\pi/\omega}\frac{dt}{T}\left(f(X,t)\frac{d^2U}{dX^2}\right)}_{\phi}$$

$$+ \underbrace{\int_0^{2\pi/\omega}\frac{dt}{T}\, f(X,t)}_{\phi} - \frac{1}{m\omega^2}\int_0^{2\pi/\omega}\frac{dt}{T}\, f(X,t)\frac{df(X,t)}{dX}$$

$$\Rightarrow \quad m\ddot{X} = -\frac{d}{dX}\left(U(X) + \frac{1}{2m\omega^2}\overline{f^2}(X,t)\right)$$

where $\overline{f^2} \equiv \int_0^{2\pi/\omega}\frac{dt}{T}\, f^2(X,t)$

In our example $\overline{f^2} = \int_0^{2\pi/\omega} \frac{dt}{T} \left(f_1 \cos\omega t + f_2 \sin\omega t\right)^2$ $\quad T = \frac{2\pi}{\omega}$

$$(u = \omega t) \quad = \int_0^{2\pi} \frac{du}{2\pi} \left(f_1 \cos u + f_2 \sin u\right)^2 = \int_0^{2\pi} \frac{du}{2\pi} \left(f_1^2 \cos^2 u + f_2^2 \sin^2 u + \cancel{2} f_1 f_2 \sin(2u)\right)$$

$$= \frac{1}{2}\left(f_1^2 + f_2^2\right)$$

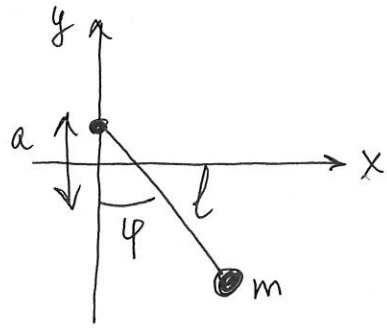
$$\Rightarrow \quad m\ddot{X} = -\frac{d}{dX}\left(U(X) + \frac{f_1^2 + f_2^2}{4m\omega^2}\right) = -\frac{d}{dX} U_{eff}(X)$$

effective potential energy

Remember that $f_{1,2}$ are functions of X!

Slow motion $X(t)$, averaged over fast oscillations $\xi(t)$, is described by the effective potential $U_{eff}(X)$.

Pendulum with oscillating support (Kapitza pendulum)



Support point coordinate

$$Y(t) = a\cos\gamma t$$

Coordinates of m:

$$x = l\sin\varphi$$

$$y = a\cos\gamma t - l\cos\varphi$$

$$K = \frac{1}{2}m(\dot{x}^2+\dot{y}^2) = \frac{1}{2}m\left[(l\cos\varphi\,\dot{\varphi})^2 + (-a\gamma\sin\gamma t + l\sin\varphi\,\dot{\varphi})^2\right]$$

$$= \frac{1}{2}m\Big[l^2\cos^2\varphi\,\dot{\varphi}^2 + l^2\sin^2\varphi\,\dot{\varphi}^2 + a^2\gamma^2\sin^2(\gamma t) - 2la\gamma\sin(\gamma t)\sin(\varphi)\,\dot{\varphi}\Big]$$

$$U = mgy$$

$$L = K - U = \frac{1}{2}m\left[l^2\dot{\varphi}^2 - 2la\gamma\sin(\gamma t)\sin\varphi\,\dot{\varphi}\right] + mgl\cos\varphi$$

where I dropped $\frac{1}{2}ma^2\gamma^2\sin^2(\gamma t)$ and $mga\cos\gamma t$ because these terms do not affect coordinate φ.

Observe that $\sin(\gamma t)\sin\varphi\,\dot{\varphi} = \frac{d}{dt}\left(\sin(\gamma t)\cos\varphi\right) - \gamma\cos(\gamma t)\cos\varphi$

Full time derivative does not affect motion because

$$\int_{t_1}^{t_2} dt\left(L_0 + \frac{dN}{dt}\right) = \int_{t_1}^{t_2} dt\,L_0 + \underbrace{N(t_2) - N(t_1)}_{\text{constant!}}$$

Therefore $L = \frac{1}{2} m \left[l^2 \dot{\varphi}^2 + 2 l a \gamma^2 \cos(\gamma t) \cos\varphi \right] + mgl\cos\varphi$

EL equ: $\frac{d}{dt} \frac{\partial L}{\partial \dot{\varphi}} = \frac{\partial L}{\partial \varphi}$

$$m l^2 \ddot{\varphi} = \underbrace{-mgl\sin\varphi}_{-\frac{dU}{d\varphi}} \underbrace{- m l a \gamma^2 \cos(\gamma t) \sin\varphi}_{f(\varphi, t) = f_1 \cos(\gamma t)}$$

Hence

let now $\varphi = \phi + \xi$

$\Rightarrow$ effective potential (obtained after averaging over fast oscillations with frequency γ)

$$U_{eff}(\phi) = U(\phi) + \frac{f_1^2}{4 m l^2 \gamma^2} =$$

$$= -mgl\cos\phi + \frac{(m l a \gamma^2 \sin\phi)^2}{4 m l^2 \gamma^2}$$

$$= -mgl\cos\phi + \frac{1}{4} m a^2 \gamma^2 \sin^2\phi$$

$$= +mgl\left(-\cos\phi + \frac{a^2\gamma^2}{4gl}\sin^2\phi\right)$$

Minima of $U_{eff}(\phi)$ describe point of stable equilibrium.

Find them: $\frac{dU_{eff}}{d\varphi} = 0 = \frac{d}{d\varphi}\left(-\cos\phi + \lambda \sin^2\phi\right) \quad \left(\lambda = \frac{a^2\gamma^2}{4gl}\right)$

$$\sin\phi + 2\lambda \sin\phi \cos\phi = 0$$

Extrema

(1) $\sin\phi = 0 \Rightarrow \phi = 0$ or π.

(2) $\cos\phi = -\frac{1}{2\lambda} = -\frac{2gl}{a^2\gamma^2}$

Minima: $\dfrac{d^2 U_{eff}}{d\varphi^2} > 0$

$$\frac{d^2 U_{eff}}{d\varphi^2} = \frac{d}{d\varphi}\left(\sin\varphi + \lambda \sin 2\varphi\right) = \cos\varphi + 2\lambda\cos 2\varphi$$

(1) $\varphi = 0 \Rightarrow U''_{eff} = 1 + 2\lambda = 1 + \dfrac{a^2\gamma^2}{2gl} > 0$ <u>YES</u>

$\varphi = \pi \Rightarrow U''_{eff} = -1 + 2\lambda = \dfrac{a^2\gamma^2}{2gl} - 1 > 0$ <u>YES</u>

if $a^2\gamma^2 > 2gl$!

For high frequency γ minimum at $\varphi = \pi$ becomes stable!!!

(2) $\cos\varphi = -\dfrac{1}{2\lambda}$, $U''_{eff} = -\dfrac{1}{2\lambda} + 2\lambda\left(2\cos^2\varphi - 1\right) =$

$$= -\frac{1}{2\lambda} + 2\lambda\left(\frac{2}{(2\lambda)^2} - 1\right) =$$

$$= -\frac{1}{2\lambda} + \frac{1}{\lambda} - 2\lambda = \frac{1}{2\lambda} - 2\lambda =$$

$$= \frac{1 - 4\lambda^2}{2\lambda} > 0 \text{ for } 2\lambda < 1.$$

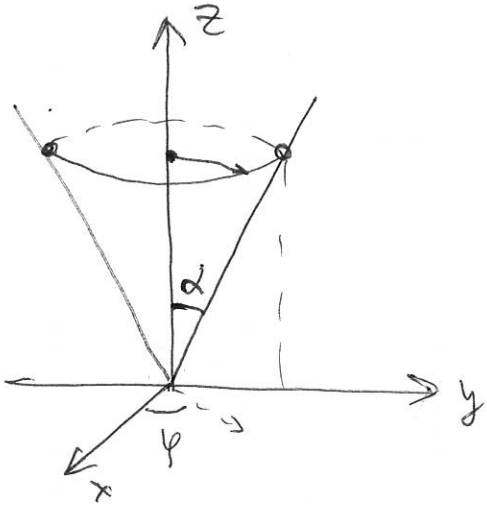
and $\lambda > 0$.

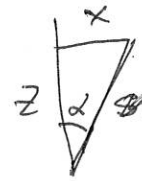
But $\cos\varphi = \dfrac{-1}{2\lambda}$ requires $|2\lambda| > 1$.

$\Rightarrow$ no minimum at $\cos\varphi = -\dfrac{1}{2\lambda}$.

Central field

Particle in a cone (cone angle 2α)





$z = s\cos\alpha$

$x = s\sin\alpha$

$$\frac{x}{z} = \mathrm{tg}\,\alpha$$

$$x = z\,\mathrm{tg}\,\alpha\cos\varphi$$

$$y = z\,\mathrm{tg}\,\alpha\sin\varphi$$

$$\dot{x} = \mathrm{tg}\,\alpha\left(\dot{z}\cos\varphi - z\sin\varphi\,\dot{\varphi}\right)$$

$$\dot{y} = \mathrm{tg}\,\alpha\left(\dot{z}\sin\varphi + z\cos\varphi\,\dot{\varphi}\right)$$

$$T = \frac{1}{2}m\left[\mathrm{tg}^2\alpha\left(\dot{z}^2 + z^2\dot{\varphi}^2\right) + \dot{z}^2\right] =$$

$$= \frac{1}{2}m\left(\frac{\dot{z}^2}{\cos^2\alpha} + \mathrm{tg}^2\alpha\, z^2\dot{\varphi}^2\right)$$

$$\mathrm{tg}^2\alpha + 1 = \frac{\sin^2\alpha + \cos^2\alpha}{\cos^2\alpha}$$

$$U = mgz \quad , \quad L = T - U$$

$\Rightarrow$ Coordinates (z, φ)

$$\frac{\partial L}{\partial \varphi} = 0 \Rightarrow p_\varphi = \frac{\partial L}{\partial \dot{\varphi}} \text{ is conserved!}$$

(1) $$\frac{\partial L}{\partial \dot{\varphi}} = m\,\mathrm{tg}^2\alpha\, z^2\dot{\varphi} = \text{const} \Rightarrow z^2\dot{\varphi} = z_0^2\dot{\varphi}_0$$

$$\frac{\partial L}{\partial \dot{z}} = \frac{m\dot{z}}{\cos^2\alpha} \qquad \frac{\partial L}{\partial z} = m\,\mathrm{tg}^2\alpha\,\dot{\varphi}^2 z - mg$$

(2) $$\frac{m\ddot{z}}{\cos^2\alpha} = -mg + m\,tg^2\alpha\,\dot{\varphi}^2 z$$

$$\ddot{z} = -g\cos^2\alpha + \sin^2\alpha\, z\,\dot{\varphi}^2$$

Conserved momentum $$p_\varphi = m\,tg^2\alpha\, z^2\dot{\varphi} \Rightarrow \dot{\varphi} = \frac{p_\varphi}{m\,tg^2\alpha\, z^2}\ ;\ p_\varphi = m\,tg^2\alpha\, z_0^2\dot{\varphi}_0$$

$$\ddot{z} = -g\cos^2\alpha + z\sin^2\alpha\,\frac{p_\varphi^2}{m^2 tg^4\alpha\, z^4} =$$

!

$$= -g\cos^2\alpha + \frac{p_\varphi^2\cos^4\alpha}{m^2\sin^2\alpha}\,\frac{1}{z^3}$$

$$= -\frac{\partial}{\partial z}\left(g\cos^2\alpha\, z + \frac{p_\varphi^2\cos^4\alpha}{2m^2\sin^2\alpha}\,\frac{1}{z^2}\right)$$

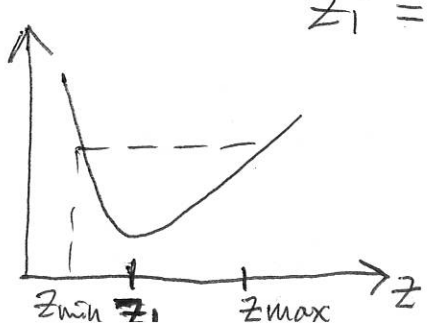
$$m\ddot{z} = -\frac{\partial}{\partial z}U_{eff} \quad (3)$$

where effective potential $U_{eff} = mg\cos^2\alpha\, z + \frac{p_\varphi^2\cos^4\alpha}{2m\sin^2\alpha}\,\frac{1}{z^2}$

Minimum of $U(z)$:

$$0 = \frac{\partial U_{eff}}{\partial z} = mg\cos^2\alpha - \frac{p_\varphi^2\cos^4\alpha}{m\sin^2\alpha}\,\frac{1}{z_1^3}$$

$$z_1^3 = \frac{p_\varphi^2\cos^2\alpha}{m^2 g\sin^2\alpha}\ ,\ z_1 = \left(\frac{p_\varphi^2}{m^2 g\, tg^2\alpha}\right)^{1/3}$$



Multiply (3) by $\dot{z}$: $\dot{z}\ddot{z} = \frac{d}{dt}\left(\frac{1}{2}\dot{z}^2\right)$

$$\Rightarrow \frac{d}{dt}\left(\frac{m}{2}(\dot{z})^2\right) + mg\cos^2\alpha\,\dot{z} - \frac{p_\varphi^2\cos^4\alpha}{m^2\sin^2\alpha}\frac{\dot{z}}{z^3} = 0$$

$$\frac{\dot{z}}{z^3} = \frac{d}{dt}\left(-\frac{1}{2z^2}\right)$$

$$\Rightarrow \frac{d}{dt}\left(\frac{m}{2}\dot{z}^2 + mg\cos^2\alpha\, z + \frac{p_\varphi^2\cos^4\alpha}{2m^2\sin^2\alpha}\frac{1}{z^2}\right) = 0$$

Energy is also conserved!

$$E = \frac{1}{2}m\dot{z}^2 + mg\cos^2\alpha\, z + \frac{p_\varphi^2\cos^4\alpha}{2m^2\sin^2\alpha}\frac{1}{z^2}$$

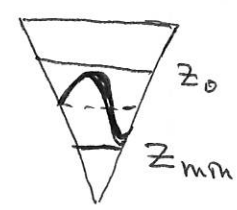
find period from this!

Lowest z_{min} which particle can reach:

$$\dot{z} = 0 \Rightarrow mg\cos^2\alpha\, z_{min} + \frac{p_\varphi^2\cos^4\alpha}{2m^2\sin^2\alpha}\frac{1}{z_{min}^2} = E$$

follows from the same equation

Highest z_{max} ~~= initial~~.



Expand about z_1: $z = z_1 + h$

$$\ddot{h} = -g\cos^2\alpha + \frac{p_\varphi^2\cos^4\alpha}{m^2\sin^2\alpha}\frac{1}{(z_1+h)^3} = -g\cos^2\alpha + \frac{p_\varphi^2\cos^4\alpha}{m^2\sin^2\alpha\, z_1^3}$$

$$(z_1+h)^{-3} = z_1^{-3}\left(1+\frac{h}{z_1}\right)^{-3} = z_1^{-3}\left(1 - \frac{3h}{z_1}\right) \quad - \frac{3p_\varphi^2\cos^4\alpha}{m^2\sin^2\alpha\, z_1^4}h$$

$\Rightarrow$ Oscillations $\ddot{h} = -\Omega^2 h$, $h(t) = A\cos(\Omega t + \phi)$

$$\Omega^2 = \frac{3p_\varphi^2\cos^4\alpha}{m^2\sin^2\alpha\, z_1^4} = \frac{3m^2 tg^4\alpha\, z_0^4\dot{\varphi}_0^2}{m^2\sin^2\alpha\, z_1^4}\cos^4\alpha = 3\sin^2\alpha\left(\frac{z_0}{z_1}\right)^4\dot{\varphi}_0^2$$

$$z^2 \dot{\varphi} = \frac{p_\varphi}{m\, tg^2\alpha}$$

$\Rightarrow$ in the same approximation $z = z_1 + h(t)$

$$(z_1 + h(t))^2\, \dot{\varphi}(t) = \frac{p_\varphi}{m\, tg^2\alpha}$$

$$\dot{\varphi}(t) = \frac{p_\varphi}{m\, tg^2\alpha} (z_1 + h)^{-2} =$$

$$= \frac{p_\varphi}{m\, tg^2\alpha} \frac{1}{z_1^2 \left(1 + \frac{h}{z_1}\right)^2} = \frac{p_\varphi}{m\, tg^2\alpha\, z_1^2} \left(1 - \frac{2h(t)}{z_1}\right)$$

$\Rightarrow$ angular velocity $\dot{\varphi}(t) \equiv \omega(t)$ is a periodic function of time!

$$\dot{\varphi}(t) = \frac{p_\varphi}{m\, tg^2\alpha\, z_1^2} \left[1 - \frac{2}{z_1} A \cos(\Omega t + \beta)\right]$$

where $h(t) = A \cos(\Omega t + \beta)$